

Johnson-Trotter

```
#include <stdio.h>
```

```
int LR = 1, RL = 0;
```

```
void swap (int *a, int *b)
```

```
{
```

```
    int temp = *a;
```

```
    *a = *b;
```

```
    *b = temp;
```

```
}
```

```
int fact (int n)
```

```
{
```

```
    int res = 1;
```

```
    for (int i = 1; i <= n; i++)
```

```
        res = res * i;
```

```
    return res;
```

```
}
```

```
int Permutation (int a[], int dir[], int n)
```

```
{
```

```
    int prev = 0, mobile = 0, pos;
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        if ((dir[pos] - 1) == -RL && i != 0)
```

```
        {
```

```
            if (a[i] > a[i-1] && a[i] > prev)
```

```
            {
```

```
                mobile = a[i];
```

```
                prev = mobile;
```

```
            }
```

```
        }
```

```

3
if (dir[a[i]-1] == LR && i != n-1)
{

```

```

    if (a[i] > a[i+1] && a[i] > prev)
    {

```

```

        mobile = a[i];

```

```

        prev = mobile;
    }
}

```

```

for (int i = 0; i < n; i++)

```

```

    if (a[i] == mobile)

```

```

        pos = i+1;

```

```

    if (dir[pos-1] == RL)

```

```

        swap(a[pos-1], a[pos-2]);

```

```

    else if (dir[pos-1] == LR)

```

```

        swap(a[pos], a[pos-1]);

```

```

for (int i = 0; i < n; i++)
{

```

```

    if (a[i] > mobile)
    {

```

```

        if (dir[a[i]-1] == LR)

```

```

            dir[a[i]-1] = RL;

```

```

        else if (dir[a[i]-1] == RL)

```

```

            dir[a[i]-1] = LR;
        }
    }

```

```

for (int i = 0; i < n; i++)

```

```

    printf("%d ", a[i]);

```

```

    printf("\n");
}

```

int main()

{

int n, a[10], dir[10]

printf("Enter the number of digits needed: ");
scanf("%d", &n);

for (int i=0; i<n; i++)

{

a[i] = i+1;

printf("%d", a[i]);

}

printf("\n");

for (int i=0; i<n; i++)

dir[i] = RL;

for (int i=1; i<fact(n); i++)

{

Permutation(a, dir, n);

return 0;

}

}

Output:

~~23/03/26~~

Enter the value of n: 3

123

132

312

321

231